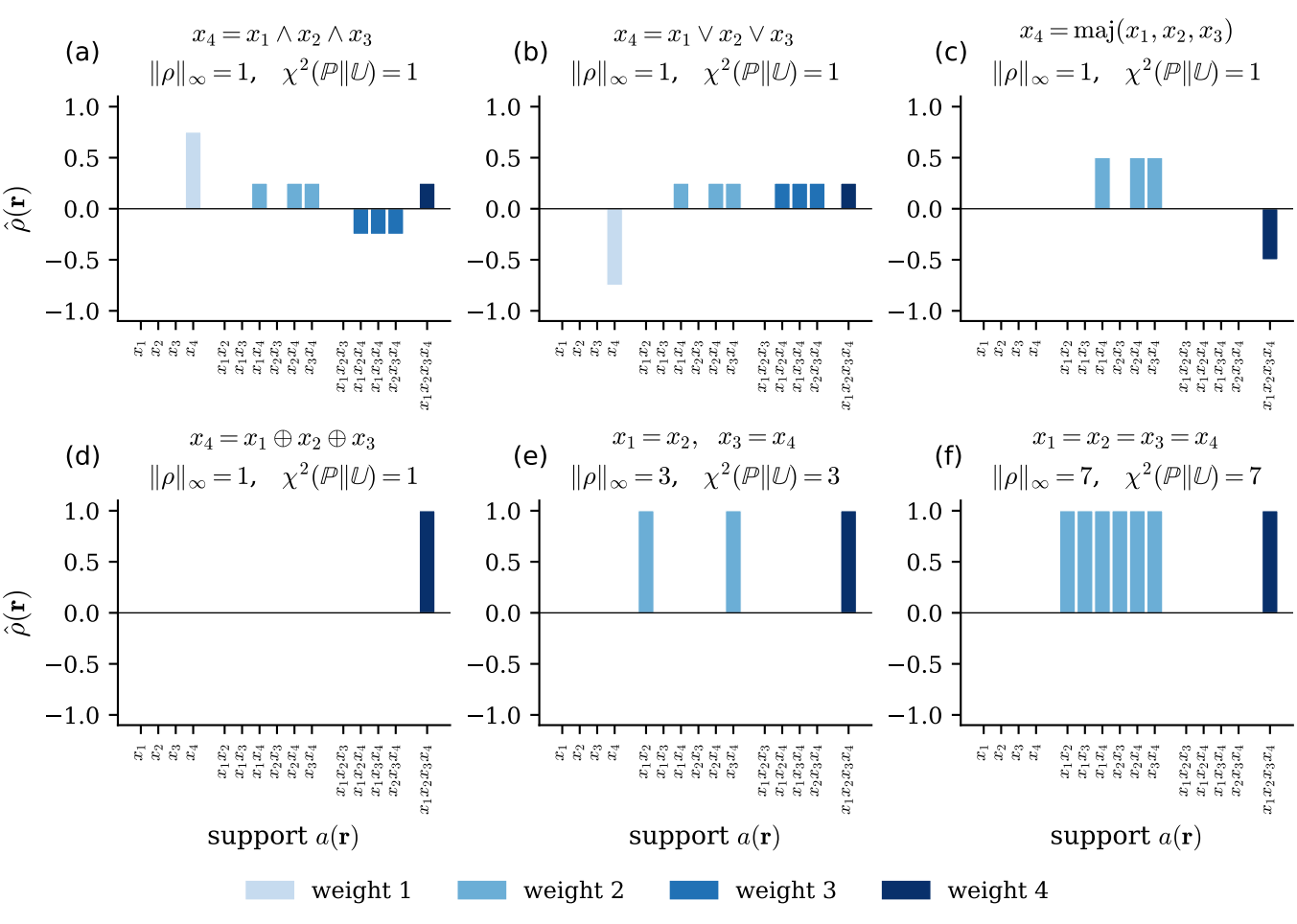
(a)
$x_4 = x_1 \wedge x_2 \wedge x_3$
$\|\rho\|_\infty = 1, \quad \chi^2(\mathbb{P}\|\mathbb{U}) = 1$
(b)
$x_4 = x_1 \vee x_2 \vee x_3$
$\|\rho\|_\infty = 1, \quad \chi^2(\mathbb{P}\|\mathbb{U}) = 1$
(c)
$x_4 = \mathrm{maj}(x_1, x_2, x_3)$
$\|\rho\|_\infty = 1, \quad \chi^2(\mathbb{P}\|\mathbb{U}) = 1$
(d)
$x_4 = x_1 \oplus x_2 \oplus x_3$
$\|\rho\|_\infty = 1, \quad \chi^2(\mathbb{P}\|\mathbb{U}) = 1$
(e)
$x_1 = x_2, \quad x_3 = x_4$
$\|\rho\|_\infty = 3, \quad \chi^2(\mathbb{P}\|\mathbb{U}) = 3$
(f)
$x_1 = x_2 = x_3 = x_4$
$\|\rho\|_\infty = 7, \quad \chi^2(\mathbb{P}\|\mathbb{U}) = 7$
$\hat{\rho}(\mathbf{r})$
1.0
0.5
0.0
−0.5
−1.0
$x_1$ $x_2$ $x_3$ $x_4$ $x_1x_2$ $x_1x_3$ $x_1x_4$ $x_2x_3$ $x_2x_4$ $x_3x_4$ $x_1x_2x_3$ $x_1x_2x_4$ $x_1x_3x_4$ $x_2x_3x_4$ $x_1x_2x_3x_4$
support $a(\mathbf{r})$
weight 1
weight 2
weight 3
weight 4